B.Tech (ME)– IV-I Sem

23A03703a	NON CONVENTIONAL ENERGY SOURCES (Professional Elective-V)	L	T	P	C
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Course objectives: The objectives of the course are to	
1	**To introduce** the principles of solar radiation and its measurement for energy applications.
2	**To explain** the types and working principles of solar energy collectors and storage systems.
3	**To provide** an understanding of various renewable energy sources including wind, biomass, and geothermal energy.
4	**To explore** ocean energy technologies and direct energy conversion systems like thermoelectric generators and fuel cells.
5	**To assess** the economic and environmental impacts of renewable energy systems and their integration into energy infrastructure.

Course Outcomes: On successful completion of the course, the student will be able to,		
1	Define and explain the fundamentals of solar radiation and analyze data using appropriate instruments and models.	L1,L2,L4
2	Compare different types of solar collectors and evaluate their thermal performance for various applications.	L3,L4,L5
3	Describe and apply renewable energy technologies such as solar heating, wind turbines, and biomass digesters to evaluate the renewable energy.	L2,L3, L5
4	Illustrate the principles and potential of ocean, geothermal, and mini-hydel energy to apply and create sustainable energy models.	L2,L3,L6
5	Analyze the principles and performance of direct energy conversion systems including fuel cells and MHD generators.	L1,L4,L5

UNIT – I

PRINCIPLES OF SOLAR RADIATION: Role and potential of new and renewable source, the solar energy option, Environmental impact of solar power, physics of the sun, the solar constant, extra-terrestrial and terrestrial solar radiation, solar radiation on titled surface, instruments for measuring solar radiation and sun shine, solar radiation data.

UNIT-II

SOLAR ENERGY COLLECTION: Flat plate and concentrating collectors, classification of concentrating collectors, orientation and thermal analysis, advanced collectors.

UNIT-III

SOLAR ENERGY STORAGE AND APPLICATIONS: Different methods, Sensible, latent heat and stratified storage, solar ponds. Solar Applications- solar heating/cooling technique, solar distillation and drying, photovoltaic energy conversion.

WIND ENERGY: Sources and potentials, horizontal and vertical axis windmills, performance characteristics, Betz criteria

UNIT-IV

BIO-MASS: Principles of Bio-Conversion, Anaerobic/aerobic digestion, types of Bio-gas digesters, gas yield, combustion characteristics of bio-gas, utilization for cooking, I.C.Engine operation and economic aspects.

GEOTHERMAL ENERGY: Resources, types of wells, methods of harnessing the energy, potential in India.

UNIT-V

OCEAN ENERGY: OTEC, Principles utilization, setting of OTEC plants, thermodynamic cycles. Tidaland wave energy: Potential and conversion techniques, mini-hydel power plants, and their economics.

DIRECT ENERGY CONVERSION: Need for DEC, Carnot cycle, limitations, and principles of DEC.

Thermo-electric generators, Seebeck, Peltier and Joule Thomson effects, Figure of merit, materials, applications, MHD generators, principles, dissociation and ionization, hall effect, magnetic flux, MHD accelerator, MHD Engine, power generation systems, electron gas dynamic conversion, economic aspects. Fuel cells, principles, faraday's law's, thermodynamic aspects, selection of fuels and operating conditions.

TEXT BOOKS:

1. Non-Conventional Energy Sources /G.D. Rai Khanna Publisher (
2. Energy Resources Utilization and Technologies, Anjaneyulu Yerramilli, Francis Tuluri, BS Publications, 2012

REFERENCES :

1. Renewable Energy Sources/ Twidell & Weir 3rd edition Routledge publisher (2015)
2. Non Conventional Energy Resources, B.H.Khan, McGrawHIll, 2015
3. Solar Power Engineering/B.S.Magal Frank Kreith & J.F.Kreith.
4. Principles of Solar Energy/ Frank Krieth & John F Kreider.
5. Non-Conventional Energy/ Ashok V Desai/ Wiley Eastern

Online Learning Resources:

https://youtube.com/playlist?list=PL3QMEfkolRFbGhXveCE7RFDBgY0_gRxkh&si=ZYwAnUNtmIwsIq

https://youtube.com/playlist?list=PLfxYQ3zfSrafm79OhjA7hvdCgm29EKrcq&si=wBQZsw0JtePVc2f-

https://youtube.com/playlist?list=PLcWcvGXrBFeRV7f9oyuuX9RabYUuanGrK&si=TVHlSydk9cPw5MsV